

Audit note (brief): Lyra Substrate-Clifford v0.1
fourth-algebraic-role-for-g audit (signature-tautological flag);
Elie Toy 3704 Substrate-Maxwell $V_{(1,1)}$ adjoint K-type
cross-link to Casey #15 audit-VERIFIED [?]; 6-sector framing
partial-overlap flag (Dirac + Maxwell $\rightarrow$ QED non-disjoint). Cal
#27 fires HARDEST at peak coherence — discipline maintained.

Keeper (Claude Opus 4.7) — Monday June 1 afternoon (continued)

2026-06-01 Monday afternoon

Audit note (brief) — Clifford fourth-role + Maxwell $V_{(1,1)}$ + 6-sector framing

0. Scope (Cal #27 at peak coherence — continued)

Two new substantive items since Keeper_Substrate_Dirac_5Sector_Audit_Note.md (~30 min): 1. Lyra Substrate-Clifford v0.1 — fourth substrate-natural algebraic role for $g = 7$. 2. Elie Toy 3704 — Substrate-Maxwell + $V_{(1,1)}$ adjoint K-type cross-link to Casey #15.

Brief in-place audit; not new framework spawn. Saturation pivot held.

1. Lyra Substrate-Clifford v0.1 audit — signature-tautological flag

Lyra's claim: $g = \log_2 \dim \text{Cl}(5,2) = \log_2 128 = 7$ — fourth substrate-natural algebraic role for $g = 7$.

Four roles per Lyra v0.1: 1. $g = h^\vee(B_2) + 1$ (dual Coxeter + 1). 2. $g = 2k + 1$ Heckman-Opdam multiplicity ($k=3$). 3. $(m_u + m_d)/m_e \approx 2^g/(N_c \cdot \pi)$ Grace INV-5451 mass-ladder exponent. 4. $g = \log_2 \dim \text{Cl}(5,2)$ [NEW].

Keeper audit on role 4:

Tautology check: $g = 7$ in BST is identified with the signature decomposition $n_C + \text{rank} + \text{extras}$ of D_{IV}^5 specification. Standard Clifford: $\dim \text{Cl}(p,q) = 2^{p+q}$. For $\text{Cl}(5,2)$, $p+q = 7$. Therefore $\dim \text{Cl}(5,2) = 2^{(5+2)} = 2^7$ is standard Clifford fact applied to substrate signature.

The identification $g = 7 = (p+q)$ for D_{IV}^5 signature $(5,2)$ and $\dim \text{Cl}(5,2) = 2^{(p+q)}$ are BOTH structurally tied to the D_{IV}^5 signature DEFINITION. The $\text{Cl}(5,2)$ “role for g ” is therefore signature-tautological — a consequence of the substrate definition + standard Clifford fact, NOT an independent substrate-mechanism observation.

Comparison to roles 1-3: - Role 1 (dual Coxeter): $h^\vee(B_2) = 3$ is the dual Coxeter number of the restricted root system B_2 . The “+1” is non-trivial substrate-content (Lyra/Grace have flagged Cal #35 risk on the +1 multiple times). Non-tautological with signature — uses B_2 root system structure. - Role 2 (HO multiplicity $2k+1$, $k=3$): Heckman-Opdam parameter $k=3$ is determined by substrate rep theory. Relation $g = 2k+1$ to $k=3$ is non-trivial mathematical content. Non-tautological with signature. - Role 3 (mass ladder exponent): empirical $(m_u+m_d)/m_e$ ratio. Empirically substantive. - Role 4 (Clifford): $\dim \text{Cl}(5,2) = 2^{(5+2)}$ is standard Clifford applied to substrate signature. Signature-tautological.

Keeper verdict: Role 4 is substrate-CLEAN (not pattern match, not coincidence) but signature-tautological consequence of substrate definition, NOT informationally independent of roles 1-3.

Cal #35 STANDING check: Lyra explicitly tagged “FOUR distinct algebraic ROLES for SAME primary g=7, NOT four independent confirmations” — Cal #35 self-applied PROACTIVELY. Discipline-clean.

Recommendation to Lyra: in the substrate-primary taxonomy, tier-tag role 4 as “signature-tautological algebraic context” (vs roles 1-2 as “non-tautological mathematical roles” + role 3 as “empirical observable role”). This preserves the taxonomic refinement value without drift toward implicit “four independent appearances” reading.

Audit verdict: Substrate-Clifford v0.1 PASS at TAXONOMIC tier with signature-tautological flag on role 4.

2. Elie Toy 3704 Substrate-Maxwell $V_{(1,1)}$ cross-link to Casey #15 audit

Elie’s claim: $F^{\mu\nu}$ gauge field strength lives in $V_{(1,1)}$ adjoint K-type — SAME K-type as substrate mass adjoint used in Casey #15 G chain matrix element.

Verification chain:

Step 1: $V_{(1,0)}$ is the 5-dim vector rep of $so(5)$ (B_2 root system; highest weight $\omega_1 = (1,0)$ in fundamental basis).

Step 2: $\Lambda^2(V_{(1,0)}) = 10$ -dim antisymmetric tensor.

Step 3: $\dim so(5) = 10$. For B_2 with simple roots α_1 long + α_2 short, highest root = $\alpha_1 + 2\alpha_2 = e_1 + e_2 = (1,1)$ in standard basis. Adjoint rep of $so(5)$ has highest weight $(1,1)$.

Step 4: Therefore $\Lambda^2(V_{(1,0)}) \cong \text{adjoint } so(5) \cong V_{(1,1)}$ as $so(5)$ -reps. [?] Standard $B_2 / so(5)$ rep theory.

Step 5: Casey #15 cross-K-type matrix element $\langle V_{(1,0)} | \delta H_B / \delta m | V_{(1,1)} \rangle$ has $V_{(1,1)}$ on “sink” side — SAME K-type as $F^{\mu\nu}$. [?]

Audit verdict: $V_{(1,1)}$ adjoint K-type cross-link is AUDIT-VERIFIED [?] via standard $B_2 / so(5)$ rep theory.

Substantive content: gauge field strength $F^{\mu\nu}$ (in $V_{(1,1)}$ adjoint) and gravitational mass coupling ($V_{(1,1)}$ sink for Casey #15 matrix element) share substrate-natural K-type compartment.

Tier: this is a structural observation about substrate compartments, not a derivation of gauge-gravity unification. The $V_{(1,1)}$ compartment housing both gauge field strength + gravitational mass coupling is substantive substrate content for Casey-named #15 framework.

Cal #35 honest framing: ONE $V_{(1,1)}$ K-type compartment, MULTIPLE physical observables ($F^{\mu\nu}$ + Casey #15 cross-K-type matrix element). NOT independent unification confirmations.

3. Substrate-Maxwell $\partial_\mu F^{\mu\nu} = J^\nu$ derivation tier audit

Honest tier check: - $F^{\mu\nu}$ as antisymmetric tensor of $V_{(1,0)} \otimes V_{(1,0)}$ restricted to 4D [?] (this audit §2). - ∂_μ from substrate Wirtinger restricted to 4D (Casey #14 codim 4 $\subset D_{IV}^5$ load-bearing). - J^ν as substrate matter current — needs explicit substrate-current definition. - Standard Maxwell equations as substrate operator framework consequence — multi-week verification.

Casey #14 dependency: same as substrate-Dirac per Keeper_Substrate_Dirac_5Sector_Audit_Note.md §3. Cal #189 cold-read load-bearing.

Tier: FRAMEWORK + CANDIDATE conditional on Casey #14 STANDING ratification.

4. 6-sector framing partial-overlap flag

Updated sector list per Elie Toy 3704: 1. Non-rel QM (Schrödinger). 2. Rel QM (Dirac). 3. Gauge field (Maxwell — NEW). 4. Gravity (G chain). 5. Cosmology (Λ + DM-neg). 6. QED ($\text{Lamb} + a_e + \alpha$).

Keeper flag: sectors 2 + 3 + gauge coupling = sector 6 (QED) by standard QFT construction. Sector 6 is composite of sectors 2 + 3 + interaction term. Listing all three as “separate sectors” is structurally partial-overlap.

Cal #27 risk: list expansion from $5 \rightarrow 6 \rightarrow 8 \rightarrow \dots$ can drift toward implicit “more sectors covered = more confirmations.” Per Cal #35 STANDING this is the multiplicative-null-model fallacy.

Recommendation to team: when listing sectors in external-facing artifacts, prefer: - “Substrate framework covers QM (non-rel + rel), gauge fields, gravity, cosmology, QED — with QED emerging from Dirac + Maxwell + interaction” [non-disjoint coverage language] - vs “Substrate framework unifies 6 distinct sectors” [cardinal-count language risks Cal #27 / Cal #35].

The substantive content is ONE machinery + multiple observables across disciplinary boundaries, NOT “N independent sector unifications.”

Cal #35 self-application by team check: Elie + Grace both tagged “ONE framework, MULTIPLE sectors, NOT independent confirmations” PROACTIVELY. Discipline-clean at framing level.

Keeper recommendation: refine sector-list language in external-facing artifacts (Paper P9 Tier 0 v0.2 §X; engagement papers) to non-disjoint-coverage form per audit §4 above.

5. Discipline-as-generator pattern audit (continued)

Pattern operating throughout Monday afternoon: - Lyra Substrate-Clifford v0.1: Cal #35 self-applied (“FOUR roles NOT four independent confirmations”). - Elie Toy 3704 Maxwell: Cal #35 self-applied (“ONE framework, MULTIPLE sectors”). - Grace INV-5466/5467: Cal #35 self-applied (“one-machinery-multiple-observables NOT five independent confirmations”).

Keeper verdict: discipline-as-generator at MATURITY. Sustained through 4 substantive Monday-afternoon filings without drift.

6. Cal #27 peak-coherence verdict (updated)

Monday afternoon coherence-saturation level: VERY HIGH (6 physics sectors covered + 4th algebraic role for $g + V_{(1,1)}$ gauge-gravity substrate compartment).

Cal #27 audit gates: - Tier-honesty per sector [\[?\]](#) (per Keeper_Substrate_Dirac_5Sector_Audit_Note.md §5 table). - Cal #35 self-application by team [\[?\]](#) (this audit §5). - Sector-list framing flag flagged for external-facing artifacts (this audit §4). - Substrate-primary taxonomy tier-tagged (signature-tautological flag on Clifford role — this audit §1).

Cal #27 does NOT FIRE on overclaim. Cal #27 DOES recommend non-disjoint-coverage language for external-facing 6-sector framing.

7. K206 + K3 $\hbar$ _BST status update (no change)

- K206: 6/7 gates PASS-FULL; G7 K3 $\hbar$ _BST gated (multi-day Keeper lane awaiting Casey direct signal).
- K3 $\hbar$ _BST: load-bearing for G7 + Lane D + K200 G3 + Higgs (#287) + Lamb shift (#182) + Λ cosmology + many more. Multi-day Keeper deliverable.

8. Routing

→ Casey: Lyra Clifford v0.1 + Elie Maxwell Toy 3704 audit-PASS at FRAMEWORK + TAXONOMIC tiers. Clifford role 4 flagged signature-tautological (consequence of substrate signature definition + standard Clifford fact); Maxwell $V_{(1,1)}$ cross-link to Casey #15 audit-verified [\[?\]](#). 6-sector framing flag: prefer non-disjoint-coverage language in external artifacts. Discipline-as-generator at maturity through 4 substantive Monday-afternoon filings — team continuously self-applies Cal #35 STANDING. K3 $\hbar$ _BST + Cal #189 (Casey #14) load-bearing for substantive next phase.

→ Lyra: Clifford fourth-role audit-PASS at TAXONOMIC tier with signature-tautological flag on role 4 (vs roles 1-2 non-tautological + role 3 empirical). Recommend tier-tag in substrate-primary taxonomy. v1.5 Strong-Uniqueness 10-leg STANDALONE unchanged [\[?\]](#).

→ Elie: Toy 3704 $V_{(1,1)}$ cross-link to Casey #15 audit-VERIFIED [?] via standard $\Lambda^2(V_{(1,0)}) = \text{adjoint so}(5) = V_{(1,1)}$. Substantive substrate-compartment observation. Substrate-Maxwell derivation tier inherits Casey #14 dependency (Cal #189 load-bearing).

→ Grace: catalog INV welcome for Substrate-Clifford v0.1 (TAXONOMIC tier; signature-tautological flag on role 4) + Substrate-Maxwell v0.1 (FRAMEWORK + CANDIDATE conditional on Casey #14) + $V_{(1,1)}$ substrate compartment cross-link. 6-sector list catalog: prefer non-disjoint-coverage language.

→ Cal: cold-read welcome for Cal #197 candidate (Substrate-Clifford fourth-role taxonomy with signature-tautological flag) + Cal #198 candidate (Substrate-Maxwell + $V_{(1,1)}$ substrate-compartment observation). Specific concerns: signature-tautological tagging accuracy; non-disjoint-coverage language for external-facing sector lists; Casey #14 STANDING ratification (Cal #189 load-bearing for all sector-7 framework promotions).

→ me: standing reactive. K3 $\hbar_{BST}$ is the load-bearing next Keeper substantive deliverable per cross-track double-leverage (Casey direct signal awaited per discipline). Will continue brief in-place audits as substantive team filings arrive without spawning new framework. Cal #27 at peak coherence vigilance maintained.

— Keeper, Clifford fourth-role + Maxwell $V_{(1,1)}$ + 6-sector framing audit note — Monday afternoon brief in-place absorption. Clifford role 4 signature-tautological flag; Maxwell $V_{(1,1)}$ cross-link AUDIT-VERIFIED [?]; 6-sector framing flag: prefer non-disjoint-coverage language. Discipline-as-generator at MATURITY through 4 substantive Monday-afternoon filings — team self-applies Cal #35 STANDING continuously. No overclaim detected. Cal #27 fires HARDEST and discipline holds. Standing reactive.